

Day 7 – Webhook Fundamentals

(Synchronous)

Setup & Prerequisites

1. Setup Checklist:
- **Virtual Environment active:** Ensure your terminal shows `(.venv)` in the prompt.
 - **Django Server running:** Run the dev server to keep configurations loaded.
 - **Webhook Test URL:** Open a browser page to `https://webhook.site` to get a temporary test URL.
 - **IDE Ready:** Open `games/webhooks.py` and `games/management/commands/check_expired_keys.py` in your IDE.
2. Prerequisites:
- You must have completed Day 6 successfully (Management command structured and working locally to expire keys).

Learning Objectives

By the end of this class, you will be able to:

- Explain the concept of Webhooks (HTTP callbacks) and contrast them with polling.
- Implement cryptographic payload signing using HMAC-SHA256.
- Enforce deterministic JSON serialization using `sort_keys=True` in `json.dumps()`.
- Use the `requests` library to make synchronous POST requests with custom timeouts and headers.
- Identify the core limitations of synchronous HTTP calls in management commands.

Classroom Timeline (45 min)

Segment	Duration	Focus
1. Hook & Big Picture	10 min	Explain the webhooks model. Contrast polling vs. event-driven callbacks.
2. Key Concepts & Core Ideas	10 min	Explain payload verification, HMAC, constant-time comparison, and JSON key sorting.

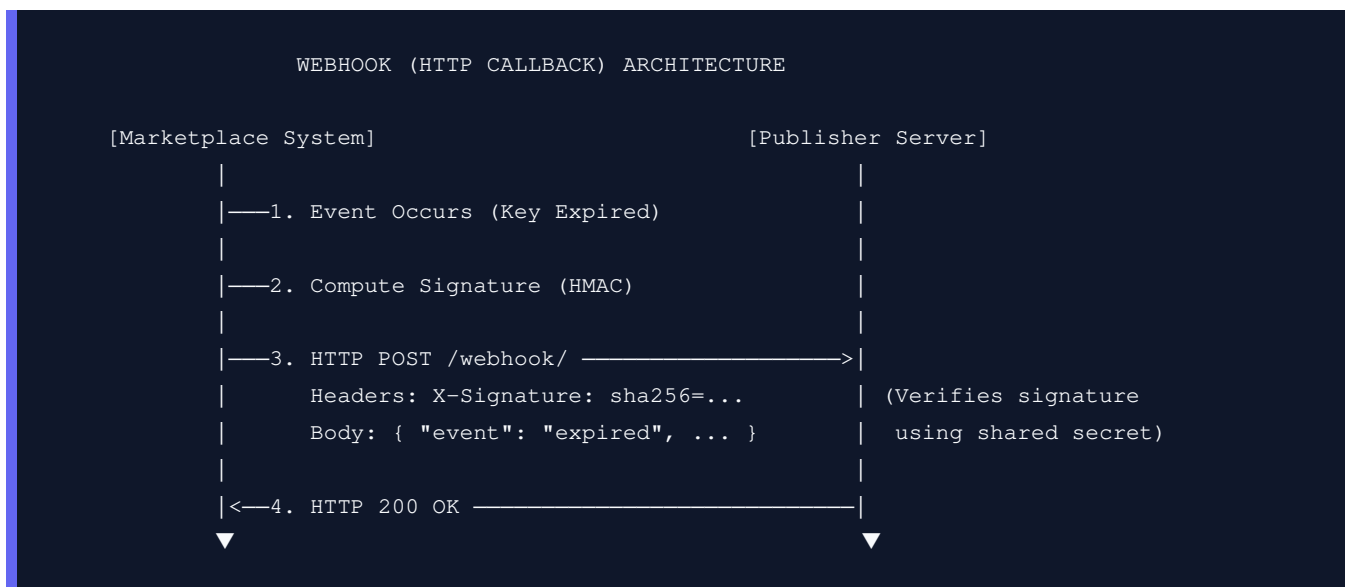
3. Live Coding Walkthrough	15 min	Write the synchronous webhook helper, integrate it with the management command, and test.
4. Discussion & Limitations	5 min	Analyze the failures of synchronous webhooks (blocking, timeout delays, no retry engine).
5. Check for Understanding	5 min	Q&A on security signatures, response codes, and sync blockages.

Lecture Step-by-Step Delivery Plan

1. Hook & Big Picture (10 min)

Polling vs Webhooks analogy: *"Imagine checking your physical mailbox every 5 minutes to see if a package arrived (polling) versus having a delivery person ring your doorbell when it arrives (webhook). Webhooks are push notifications for APIs."*

- **Security Risk:** Since any client can send a POST request to the publisher's public webhook URL, the publisher needs a way to guarantee the request actually came from our marketplace. We solve this using cryptographic payload signing.



2. Key Concepts & Core Ideas (10 min)

Introduce these webhook security and network concepts:

- **HMAC-SHA256:** Hash-based Message Authentication Code. Both systems share a secret key. We sign the payload bytes with this secret, producing a signature hex digest. The publisher recalculates the signature using the same secret and matches them.
 - **Question:** Why sign? Because anyone can POST to the publisher's webhook URL.

■ *Answer:* A signature verified with a shared secret ensures the payload is authentic, has not been tampered with in transit, and was not sent by an attacker.

- **sort_keys=True in JSON:** JSON objects are unordered maps. If keys are serialized in different orders on the sender and receiver (e.g. {"key": 1, "game": 2} VS {"game": 2, "key": 1}), the raw byte sequences will differ, causing the HMAC signatures to mismatch. Sorting keys makes serialization deterministic.
 - **Question:** Why do we pass `sort_keys=True` to `json.dumps()`?
 - *Answer:* JSON key order is not guaranteed. Sorting makes the serialized string deterministic, ensuring signature calculations match.
- **hmac.compare_digest():** A utility that compares digests in constant time. This prevents timing attacks (where attackers deduce correct signature bytes by measuring tiny variations in execution time).
- **Outbound Timeouts:** Why we must always set a timeout (e.g. `timeout=5`) when calling external services. Without a timeout, a hanging remote server will cause our command line process to block indefinitely.
 - **Question:** Why do we set a timeout on requests?
 - *Answer:* Without a timeout, a hanging remote server will cause our script to block indefinitely, consuming system resources and delaying other tasks.
- **raise_for_status():** A method that raises an HTTP error if the server returns a 4xx or 5xx error code.

3. Live Coding Walkthrough (15 min)

Step 3.1: Implementing the Webhook Sender

- **Concept:** Create `games/webhooks.py`. We will write a helper function that constructs the expiry event payload, signs it using HMAC-SHA256, sets the custom headers, and dispatches the synchronous POST request.
- **Code:** Create `games/webhooks.py`:

```
import hmac
import hashlib
import json
import requests

def send_expiry_webhook(publisher, game_key_str, game_title, expired_at):
    # Define the JSON webhook payload data structure
    payload = {
        "event": "game_key.expired",
        "game_key": game_key_str,
        "game_title": game_title,
        "expired_at": expired_at.isoformat(),
    }

    # Secrets must be in bytes for cryptographic operations
```

```

secret = publisher.webhook_secret.encode('utf-8')

# Use sort_keys=True to guarantee consistent, deterministic string ordering before encoding
body = json.dumps(payload, sort_keys=True).encode('utf-8')

# Compute the HMAC SHA-256 signature using the secret key and encoded body bytes
signature = hmac.new(secret, body, hashlib.sha256).hexdigest()

headers = {
    "Content-Type": "application/json",
    # Standard header format: send signature with sha256= prefix
    "X-Signature": f"sha256={signature}",
}

try:
    # Dispatch synchronous POST request. timeout=5 prevents hanging indefinitely
    response = requests.post(publisher.webhook_url, json=payload, headers=headers, timeout=5)

    # Raise an exception if the response contains a 4xx or 5xx error status code
    response.raise_for_status()
except Exception as e:
    print(f"[Webhook] Delivery failed for publisher {publisher.id}: {e}")

```

- **Verify:** Check that the module imports compile without syntax errors.
- **⚠ Common Pitfalls:**
 - **Type errors with secrets:** Note that `hmac.new()` requires bytes, not strings. Both the `secret` and the `body` must be `.encode()`ed to bytes before hashing.

Step 3.2: Updating the Management Command

- **Concept:** Modify `check_expired_keys.py` to trigger the synchronous webhook sender for every key we mark as expired.
- **Code:** Update `games/management/commands/check_expired_keys.py`:

```

from django.core.management.base import BaseCommand
from django.utils import timezone
from games.models import GameKey
from games.webhooks import send_expiry_webhook # Import webhook sender helper

class Command(BaseCommand):
    help = 'Mark expired keys and notify publishers synchronously.'

    def handle(self, *args, **options):
        # select_related performs SQL JOINS to fetch publisher details in a single query
        expired_keys = GameKey.objects.select_related('game__publisher').filter(
            expires_at__lte=timezone.now(),
            status='active'
        )

        # Bulk update status
        count = expired_keys.update(status='expired')

```

```
# Loop over expired keys and send webhook to publisher
for key in GameKey.objects.filter(status='expired').select_related('game__publisher'):
    send_expiry_webhook(key.game.publisher, key.key_string, key.game.title, key.expire

self.stdout.write(f'Expired {count} keys (sync webhooks sent).')
```

- **Verify:**

- Open `https://webhook.site` in a browser and copy the unique URL.
- Create a publisher in your Django admin panel and paste this URL into their `webhook_url` field.
- Expire a key in admin (by setting `expires_at` in the past and status to `active`).
- Run `python manage.py check_expired_keys`.
- Check the `webhook.site` logs to confirm that the HTTP POST request arrived with the correct JSON body and the `X-Signature` header.

4. Discussion & Limitations (5 min)

Analyze the failures of synchronous webhooks:

1. **Thread Blocking:** The entire script blocks during the HTTP call.
 2. **Cascading Slowness:** If one publisher's server is slow, the script takes longer, delaying webhooks for all subsequent publishers.
 3. **No Resiliency (Retries):** If a publisher's server is down (connection error), the webhook is lost forever.
- *Preview: Tomorrow we will solve these three issues using an asynchronous task queue (Celery + Redis).*

5. Check for Understanding (5 min)

Question: "A publisher claims they received a webhook from us but the payload was tampered with. How does HMAC protect against this?"

Answer: HMAC uses a shared secret to generate a cryptographic signature of the request body. If any character in the payload is altered during transit, the receiver's computed signature will not match the `X-Signature` header, revealing that the payload was modified and should be rejected.

Question: "Why do we pass `sort_keys=True` to `json.dumps()`? What could go wrong without it?"

Answer: JSON keys are unordered. If we don't sort keys, the serialization format might change (e.g. dictionary keys serialized in a different order), resulting in a different byte sequence. This would produce a different HMAC signature, causing verification to fail even if the content remains identical.

Question: "What happens to the management command if the publisher's server is down and we're making synchronous HTTP calls?"

Answer: The management command thread will block, waiting for the HTTP request to timeout (e.g. 5 seconds per request). If many keys expired for that publisher, or multiple publishers are down, the script will take a very long time to complete and could cause other pending updates to stall.

6. Daily Checkpoint & Wrap-up (3 min)

- **Verification Criteria:**

- ☐ The `send_expiry_webhook` function is implemented in `games/webhooks.py`.
- ☐ Running `python manage.py check_expired_keys` dispatches an HTTP POST request to the publisher's webhook URL.
- ☐ The HTTP request header contains a valid `x-Signature` starting with `sha256=`.
- ☐ The webhook payload arrives at `webhook.site` successfully.